Species-Aware Review: Protocatechuate *ortho*-Cleavage Pathway in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Query bucket:** KEGG `ppu00362` "Benzoate degradation" (broad overview map) **Review topic:** Protocatechuate *ortho*-cleavage pathway (the protocatechuate branch of the β-ketoadipate pathway) **Date:** 2026-08-11 · Finalized Iteration 5 of 5

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1. Executive Summary

- The **protocatechuate (PCA) *ortho*-cleavage branch of the β-ketoadipate pathway is present and biochemically complete** in *P. putida* KT2440. All six enzymatic steps (protocatechuate → β-ketoadipate → acetyl-CoA + succinyl-CoA) are encoded by high-confidence, well-characterized genes: **pcaG/pcaH, pcaB, pcaC, pcaD, pcaI/pcaJ, pcaF**. The module should be marked **covered / satisfiable**.
- The provided 40-gene candidate list is **exactly the membership of the broad KEGG overview map `ppu00362` "Benzoate degradation"** (verified by `rest.kegg.jp/link/ppu/path:ppu00362`). It is therefore **not** a curated protocatechuate module and contains extensive **annotation over-propagation** (fatty-acid β-oxidation, phenylacetate-CoA, gallate, and generic thiolase genes).
- Only ~**9 genes are true PCA-branch members** (pcaGHBCDIJF + the entry enzyme pobA). A parallel **catechol *ortho*-cleavage branch** (catA/B/C) converges at the same lower pathway and is closely related but should be modelled as a **separate branch**.
- **KEGG has no dedicated "protocatechuate → 3-oxoadipate" module** (only M00568 "Catechol *ortho*-cleavage" exists). A new, tightly-scoped module document is recommended; the generic `ppu00362` boundary is too broad and should be flagged **module_needs_revision**.
- One clear **over-propagation** for curators: **PP_3648** is mapped by KEGG to EC 4.1.1.44 (same as pcaC/PP_1381), but InterPro shows it carries only the generic AhpD-like domain

(IPR003779) and lacks the pcaC-specific signature IPR012788 — so it should **not** satisfy the decarboxylation step. The KT2440 regulator **pcaR is PP_1375** (upstream of pcaK), pinning the previously species-transferred regulatory assignment.

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2. Target-Organism Pathway Definition

Process included. Intradiol (*ortho*) ring cleavage of protocatechuate (3,4-dihydroxybenzoate) by a non-heme Fe(III) dioxygenase, followed by lactonization, decarboxylation, hydrolysis, CoA transfer, and thiolytic cleavage, yielding the TCA-cycle intermediates **succinyl-CoA** + **acetyl-CoA**. This is the **protocatechuate branch of the β-ketoadipate pathway** — a chromosomally encoded convergent pathway widely distributed in soil bacteria and fungi (Harwood & Parales, 1996,

P 8905091

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Reaction sequence (target-organism gene → EC):

#	Reaction	Gene (locus)	EC	KEGG KO
1	protocatechuate + O_2 → 3-carboxy-*cis*,*cis*-muconate	**pcaG** (PP_4655) + **pcaH** (PP_4656)	1.13.11.3	K00448/ K00449
2	3-carboxy-*cis*,*cis*-muconate → γ-carboxymuconolactone	**pcaB** (PP_1379)	5.5.1.2	K01857
3	γ-carboxymuconolactone → β-ketoadipate enol-lactone + CO_2	**pcaC** (PP_1381)	4.1.1.44	K01607
4	β-ketoadipate enol-lactone → β-ketoadipate	**pcaD** (PP_1380)	3.1.1.24	K01055
5	β-ketoadipate + succinyl-CoA → β-ketoadipyl-CoA + succinate	**pcaI** (PP_3951) + **pcaJ** (PP_3952)	2.8.3.6	K01031/ K01032
6	β-ketoadipyl-CoA + CoA → succinyl-CoA + acetyl-CoA	**pcaF** (PP_1377)	2.3.1.174	K07823

Entry (boundary) step: pobA (PP_3537, EC 1.14.13.2, p-hydroxybenzoate 3-monooxygenase) converts 4-hydroxybenzoate → protocatechuate and is the canonical feeder reaction (Bertani et al., 2001, 11390692). Protocatechuate is also generated from quinate/shikimate, vanillate, and — after conversion — from other lignin-derived aromatics.

Alternate names / database definitions. - "β-ketoadipate pathway (protocatechuate branch)"; "PCA *ortho*-cleavage"; "protocatechuate 3,4-cleavage pathway"; the enzymes are the *pca* genes. - KEGG scatters these enzymes across maps `ppu00362` (Benzoate degradation), `ppu00360` (Phenylalanine metabolism), and the overview map `ppu01220` (Degradation of aromatic compounds). MetaCyc groups them under "protocatechuate degradation I (*meta*-cleavage is separate)".

Neighboring pathways to keep SEPARATE: - Catechol *ortho*-cleavage branch (catA/B/C → β-ketoadipate enol-lactone): parallel branch of the same β-ketoadipate pathway; converges at pcaD. KEGG module **M00568**. - **Benzoate 1,2-dioxygenase pathway** (benABCD → catechol): *upstream* of the catechol branch, KEGG map `ppu00622`. A boundary, not part of PCA cleavage. - **Gallate degradation** (galA + galBCD → 4-oxalomesaconate → pyruvate + oxaloacetate): a **type-II extradiol / 4,5-type** ring cleavage that is *mechanistically distinct from ortho-cleavage* (Nogales et al., 2005, 16030014). - **Phenylacetate-CoA (paa) pathway**, **fatty-acid β-oxidation (fad)**, and **generic β-ketothiolases** — unrelated central/peripheral pathways that share only EC-class (thiolase/hydratase/dehydrogenase) similarity.

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3. Expected Step Model (satisfiability)

Step	Expected enzyme	Status in KT2440	Gene
PCA ring cleavage	protocatechuate 3,4-dioxygenase (αβ)	**covered**	pcaG/pcaH
cycloisomerization	3-carboxy-*cis*,*cis*-muconate cycloisomerase	**covered**	pcaB
decarboxylation	γ-carboxymuconolactone decarboxylase	**covered**	pcaC
enol-lactone hydrolysis	β-ketoadipate enol-lactone hydrolase	**covered**	pcaD
CoA transfer	3-oxoadipate CoA-transferase (αβ)	**covered**	pcaI/pcaJ
thiolysis	β-ketoadipyl-CoA thiolase	**covered**	pcaF
substrate uptake	4-HB/PCA transporter; dicarboxylate permease	**covered (transport)**	pcaK (PP_1376), pcaT (PP_1378)
regulation	PcaR (IclR-family activator), induced by β-ketoadipate	**covered (regulatory)**	pcaR (PP_1375)
entry from 4-HB	p-hydroxybenzoate 3-monooxygenase	**covered (boundary)**	pobA (PP_3537)

PcaR (IclR-family activator; KT2440 locus **PP_1375**, UniProt Q88N41, verified by UniProt gene:pcaR query for taxon 160488) sits immediately upstream of pcaK, completing the cluster **pcaR(PP_1375)–pcaK(PP_1376)–pcaF(PP_1377)–pcaT(PP_1378)–pcaB(PP_1379)–pcaD(PP_1380)–pcaC(PP_1381)**. It activates pcaBDC, pcaIJF and pcaK; β-ketoadipate is the inducer, and pcaK (but not pcaF) is additionally repressed by benzoate (Nichols & Harwood, 1995,

P 8522507

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No PCA-branch enzymatic step is missing.

No steps are "not expected." The complete pathway is a defining metabolic capability of KT2440 and is directly documented experimentally in *P. putida*.

4. Candidate Genes and Evidence

4a. High-confidence PCA-branch genes (assign to module = covered)

- **pcaG (PP_4655) / pcaH (PP_4656)** — protocatechuate 3,4-dioxygenase α/β (EC 1.13.11.3, K00448/K00449). Adjacent on chromosome (verified). Deleting *pcaHG* abolishes PCA degradation in KT2440 (Dias et al., 2023, P 36357545). **Evidence: direct (target organism).**
- **pcaB (PP_1379)** — 3-carboxy-*cis,cis*-muconate cycloisomerase (EC 5.5.1.2). In pcaK-pcaF-pcaT-**pcaB**-pcaD-pcaC cluster. **Direct/strong.**
- **pcaC (PP_1381)** — γ-carboxymuconolactone (4-carboxymuconolactone) decarboxylase (EC 4.1.1.44). **Direct/strong.** *Caveat:* a paralog (PP_3648) shares the EC (see §5).
- **pcaD (PP_1380)** — 3-oxoadipate enol-lactonase (EC 3.1.1.24); annotated "enol-lactonase **2**", implying possible isozyme naming. **Direct/strong.**
- **pcaI (PP_3951) / pcaJ (PP_3952)** — 3-oxoadipate CoA-transferase α/β (EC 2.8.3.6). Stand-alone locus next to a chemotaxis transducer (consistent with β-ketoadipate chemotaxis biology). **Direct/strong.** (Note: PP_3952 curated UniProt P0A101.)
- **pcaF (PP_1377)** — β-ketoadipyl-CoA thiolase (EC 2.3.1.174, K07823). Required for **both** benzoate (catechol branch) and 4-HB (PCA branch) degradation; PcaR-regulated (Nichols & Harwood, 1995, P 8522507). **Direct/strong.**

4b. Pathway-context genes (functionally part of the module, not core cleavage enzymes)

- **pobA (PP_3537)** — 4-hydroxybenzoate 3-monooxygenase (EC 1.14.13.2). Entry step producing PCA. **Boundary — include as feeder.**
- **pcaK (PP_1376)** and **pcaT (PP_1378)** — 4-HB/PCA MFS transporter and dicarboxylate permease. **Transport steps.**
- **pcaR (PP_1375, UniProt Q88N41)** — IclR-family transcriptional activator of the pca regulon; integral regulatory member of the main cluster. **Regulatory step.**

4c. Catechol *ortho*-cleavage branch (parallel branch — model separately, converges at pcaD)

- **catA-I (PP_3713)**, **catA-II (PP_3166)** — catechol 1,2-dioxygenase (EC 1.13.11.1, K03381). Two genuine isozymes: catA-I in the *catBCA/catR* cluster; catA-II embedded in the *ben* cluster (benABCD-benK-**catA-II**-benE), co-induced with benzoate catabolism. **Redundancy is real, not over-propagation.**
- **catB (PP_3715)** — muconate cycloisomerase (EC 5.5.1.1); **catC (PP_3714)** — muconolactone Δ-isomerase (EC 5.3.3.4). Cluster with LysR regulator **catR** (PP_3716).

4d. Boundary/upstream (keep separate from PCA cleavage)

- **benA/B/C/D (PP_3161–3164)** — benzoate 1,2-dioxygenase system → cyclohexadiene-diol-carboxylate → catechol (feeds the catechol branch, not PCA). KEGG map `ppu00622`.

4e. Over-propagated / unrelated (exclude from module)

- **gal genes — galB (PP_2515), galC (PP_2514), galD (PP_2513):** gallate degradation via **4-oxalomesaconate** (EC 4.2.1.83 / 4.1.3.17 / 5.3.2.8). galA is a **type-II extradiol** dioxygenase, sharing ancestry with protocatechuate **4,5-**dioxygenase, **not** the 3,4-*ortho* enzyme (Nogales et al., 2005, P 16030014). **Distinct pathway.**
- **paa genes — paaJ (PP_3280), paaH (PP_3282), paaF (PP_3284):** phenylacetate-CoA (aerobic hybrid) pathway. Thiolase/hydratase/dehydrogenase EC overlap only.
- **fad/β-oxidation — fadA (PP_2051, PP_2137), fadB (PP_2136), PP_2217, hbd (PP_3755):** fatty-acid oxidation.
- **Generic thiolases EC 2.3.1.9/2.3.1.16 — PP_0582, PP_2215, PP_3355, bktB (PP_3754), yqeF (PP_4636):** broad acetyl-CoA acetyltransferase / β-ketothiolase family; over-propagated via EC-class similarity to pcaF.
- **gcdH (PP_0158)** glutaryl-CoA dehydrogenase; **PP_1218** acyl-CoA thioesterase; **PP_2504** 4-oxalocrotonate tautomerase (*meta*-cleavage); **PP_1791** 4-hydroxy-2-oxovalerate aldolase (*meta*-cleavage); **PP_1950** cytochrome P450; **fdx (PP_0847)** 2Fe-2S ferredoxin — none are PCA *ortho*-cleavage enzymes.

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5. Gaps, Ambiguities, and Likely Over-Annotations

1. **PP_3648 (candidate_uncertain — resolved to over-propagation).** KEGG maps it to K01607/EC 4.1.1.44, identical to pcaC (PP_1381), but **InterPro discriminates the two decisively**: pcaC (PP_1381, Q88N35) carries the *specific* signature **IPR012788 "4-carboxymuconolactone decarboxylase (PcaC)"**, whereas **PP_3648 (Q88GS0) has only the *generic* domain IPR003779 (carboxymuconolactone-decarboxylase-like / AhpD superfamily) + Pfam PF02627** and lacks IPR012788. Both share PANTHER PTHR33570:SF2 and PF02627, which is exactly why KEGG over-propagated EC 4.1.1.44 onto PP_3648. **Conclusion: PP_1381 is the true PcaC (module = covered); PP_3648 should NOT satisfy the decarboxylation step and should be flagged over-propagated/candidate_uncertain** (likely an AhpD-like redox/stress protein).
2. **pcaD naming ("enol-lactonase 2").** Suggests the annotation pipeline recognized a second enol-lactonase; verify there is no mis-split. Functionally the PP_1380 assignment is sound.
3. **Broad EC/thiolase over-propagation.** EC 2.3.1.9/2.3.1.16 thiolases (PP_0582, PP_2215, PP_3355, bktB, yqeF) and paaJ (also carrying EC 2.3.1.174) can be spuriously linked to pcaF. Only **pcaF (PP_1377, K07823)** is the β-ketoadipate-pathway thiolase; **paaJ (PP_3280)** carries 2.3.1.174 but belongs to the phenylacetate route — a real EC-overlap trap for curators.
4. **catA redundancy vs. over-propagation.** catA-I and catA-II are two authentic catechol 1,2-dioxygenases; keep both but in the **catechol** branch, not the PCA branch.
5. **No true gaps** in the PCA branch: every enzymatic step has a strong, mostly direct assignment.

Per-step paralog scan (UniProt EC search, organism 160488) — curation confidence:

Step (EC)	# KT2440 loci with this EC	Module gene	EC-unique?	Curation note
1.13.11.3	2 (α+β of one enzyme)	pcaG/pcaH	yes	single enzyme; unambiguous
5.5.1.2	1	pcaB	**yes**	single copy, high confidence
4.1.1.44	1	pcaC (PP_1381)	**yes**	PP_3648 is NOT given this EC by UniProt → KEGG-only over-propagation
3.1.1.24	1	pcaD	**yes**	single copy; "enol-lactonase 2" label does not imply a 2nd EC 3.1.1.24 gene
2.8.3.6	4	pcaI/pcaJ	no	**atoA/atoB (PP_3122/PP_3123)** = acetoacetate CoA-transferase paralog; exclude
2.3.1.174	6	pcaF	no	shared with paaJ, yqeF, PP_2215, bktB, fadA → identify pcaF by KO **K07823** + context, not EC
1.13.11.1	2	catA-I/ catA-II	(branch)	two genuine catechol-branch isozymes

Take-home: four of six enzymatic steps are EC-unique (very high confidence); only the **CoA-transferase (pcaIJ)** and **thiolase (pcaF)** steps sit in multi-member EC families and must be pinned by KEGG orthology/genomic context rather than EC number.

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6. Module and GO-Curation Recommendations

Module step status:

Module element	Recommended status
protocatechuate 3,4-dioxygenase (pcaGH)	**covered**
3-carboxymuconate cycloisomerase (pcaB)	**covered**
γ-carboxymuconolactone decarboxylase (pcaC)	**covered** (PP_1381; PP_3648 = candidate_uncertain paralog)
enol-lactone hydrolase (pcaD)	**covered**
3-oxoadipate CoA-transferase (pcaIJ)	**covered**
β-ketoadipyl-CoA thiolase (pcaF)	**covered** (exclude paaJ/generic thiolases)
PCA supply from 4-HB (pobA)	**covered (boundary/feeder)**
Whole `ppu00362` candidate list as the module	**module_needs_revision** (prune to pca + feeders)

Boundary corrections: - The generic `ppu00362` "Benzoate degradation" overview map is **too broad** to serve as this module. Split into: (a) protocatechuate *ortho*-cleavage (this module), (b) catechol *ortho*-cleavage (M00568), (c) benzoate→catechol (ben), (d) gallate/4-oxalomesaconate, (e) phenylacetate-CoA, (f) β-oxidation. Genes in (c)–(f) should not satisfy this module.

New documents / GO requests: - **Author a dedicated module** "protocatechuate *ortho*-cleavage: protocatechuate ⇒ β-ketoadipate ⇒ acetyl-CoA + succinyl-CoA" with the six EC steps above — KEGG currently lacks it (only the catechol counterpart M00568 exists). - GO annotations are largely available (GO:0018578 protocatechuate 3,4-dioxygenase activity; GO:0019619 3,4-dihydroxybenzoate catabolic process). Ensure pcaB/C/D/I/J/F carry the **process** term GO:0019619 and appropriate MF terms; request/verify a specific term for "3-oxoadipate CoA-transferase activity" (EC 2.8.3.6) if missing.

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7. Genes to Promote to Full `fetch-gene` Review

Priority order: 1. **PP_3648** — pcaC paralog ambiguity now **largely resolved by InterPro** (generic IPR003779/AhpD-like, lacks pcaC-specific IPR012788). Promote only to *confirm* it is not a functional decarboxylase and to reassign it away from EC 4.1.1.44. *High curation value (fixes an over-propagation).* 2. **pcaD (PP_1380)** — clarify the "enol-lactonase 2" designation / possible isozyme. 3. **paaJ (PP_3280)** — confirm it is excluded from the PCA branch despite carrying EC

2.3.1.174 (EC-overlap trap). 4. **catA-I (PP_3713) vs catA-II (PP_3166)** — document isozyme roles and branch assignment. 5. **pcaR (PP_1375)** — locus now **pinned** (UniProt Q88N41; upstream of pcaK). Optional: confirm the KT2440 regulon membership experimentally (transfer from *P. putida* PRS2000 is strong).

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8. Key References

- Harwood CS, Parales RE. *The β-ketoadipate pathway and the biology of self-identity.* Annu Rev Microbiol. 1996;50:553–590.

  P 8905091

  (Defines protocatechuate & catechol branches; conservation across *P. putida* and others.)
- Nichols NN, Harwood CS. *Repression of 4-hydroxybenzoate transport and degradation by benzoate…* J Bacteriol. 1995.

  P 8522507

  (pca structural genes = complete protocatechuate branch; PcaR regulation; benzoate repression of pcaK.)
- Nogales J, et al. *Molecular characterization of the gallate dioxygenase from Pseudomonas putida KT2440…* J Biol Chem. 2005.

  P 16030014

  (galA/gal pathway is a distinct type-II extradiol/4-oxalomesaconate route — not *ortho*-cleavage.)
- Dias ACF, et al. *From degrader to producer: reversing the gallic acid metabolism of Pseudomonas putida KT2440.* 2023.

  P 36357545

  (pcaHG deletion blocks PCA degradation in KT2440 — direct functional evidence.)
- Bertani I, et al. *Regulation of the p-hydroxybenzoic acid hydroxylase gene (pobA)…* 2001.

  P 11390692

  (pobA → protocatechuate → β-ketoadipate pathway.)
- Gerischer U, et al. *PcaU, a transcriptional activator of genes for protocatechuate utilization in Acinetobacter.* 1998.

P 9515921
(pcaIJFBDKCHG operon architecture — related-organism comparator; transfer to *Pseudomonas* is partial: gene set conserved, cluster organization differs.)

- KEGG (rest.kegg.jp), verified 2026-08-11: `link/ppu/path:ppu00362` (candidate list = overview-map membership); gene KO/EC for pcaGHBCDIJF, catABC, benABCD; module list (M00568 catechol *ortho*-cleavage; no protocatechuate module).

Evidence-strength notes

- **Direct (target organism, experimental):** pcaHG function (KT2440 deletion), pcaF/pcaK regulation and PcaR control (*P. putida* PRS2000/KT2440), galA distinctness (KT2440), pobA feeder role (*P. putida*).
- **Strong homology + genomic context (target genome):** pcaB, pcaC, pcaD, pcaI, pcaJ assignments (clustered, correct KO/EC).
- **Related-organism transfer:** *Acinetobacter* operon architecture (comparator only; conserved enzymes, different cluster layout — transfer of gene *identity* strong, of *organization* weak).
- **Uncertain (now resolved):** PP_3648 identity → generic AhpD-like/CMD family (IPR003779), over-propagated to EC 4.1.1.44 by KEGG only; pcaR locus → PP_1375 (UniProt Q88N41). Residual open question: whether PP_3648 has any physiological decarboxylase activity (recommend experimental check).

Appendix A — Disposition of all 40 candidate genes (curation quick-reference)

Locus	Gene	Disposition for THIS module	Rationale
PP_4655	pcaG	**covered** (core)	protocatechuate 3,4-dioxygenase α (EC 1.13.11.3, K00448)
PP_4656	pcaH	**covered** (core)	protocatechuate 3,4-dioxygenase β (EC 1.13.11.3, K00449)
PP_1379	pcaB	**covered** (core)	EC 5.5.1.2, single-copy
PP_1381	pcaC	**covered** (core)	EC 4.1.1.44, specific IPR012788
PP_1380	pcaD	**covered** (core)	EC 3.1.1.24, single-copy
PP_3951	pcaI	**covered** (core)	3-oxoadipate CoA-transferase α (EC 2.8.3.6, K01031)
PP_3952	pcaJ	**covered** (core)	3-oxoadipate CoA-transferase β (EC 2.8.3.6, K01032)
PP_1377	pcaF-I	**covered** (core)	β-ketoadipyl-CoA thiolase (EC 2.3.1.174, K07823)
PP_3537	pobA	**covered** (feeder/boundary)	4-HB 3-monooxygenase supplies PCA
PP_1376	pcaK	**covered** (transport)	4-HB/PCA MFS transporter
PP_1378	pcaT	**covered** (transport)	dicarboxylate permease
PP_1375	pcaR	**covered** (regulatory)	IclR-family activator of pca regulon
PP_3713	catA-I	separate branch (catechol)	catechol 1,2-dioxygenase isozyme
PP_3166	catA-II	separate branch (catechol)	catechol 1,2-dioxygenase isozyme (ben cluster)
PP_3715	catB	separate branch (catechol)	muconate cycloisomerase (EC 5.5.1.1)
PP_3714	catC	separate branch (catechol)	muconolactone Δ-isomerase (EC 5.3.3.4)
PP_3161–3164	benABCD	upstream/boundary	benzoate 1,2-dioxygenase → catechol

Locus	Gene	Disposition for THIS module	Rationale
PP_3648	—	**candidate_uncertain** / **over-propagated**	generic AhpD/CMD (IPR003779); not true PcaC
PP_3280	paaJ	exclude (phenylacetate)	EC 2.3.1.174 overlap only
PP_3282	paaH	exclude (phenylacetate)	3-hydroxyadipyl-CoA DH
PP_3284	paaF	exclude (phenylacetate)	enoyl-CoA hydratase-isomerase
PP_2513	galD	exclude (gallate/4-OMA)	distinct non-ortho route
PP_2514	galC	exclude (gallate/4-OMA)	CHA aldolase
PP_2515	galB	exclude (gallate/4-OMA)	OMA hydratase
PP_2051	fadA	exclude (β-oxidation)	3-ketoacyl-CoA thiolase
PP_2136	fadB	exclude (β-oxidation)	FA oxidation complex α
PP_2137	fadA	exclude (β-oxidation)	3-ketoacyl-CoA thiolase
PP_2217	—	exclude (β-oxidation)	enoyl-CoA hydratase
PP_3755	hbd	exclude (β-oxidation)	3-hydroxybutyryl-CoA DH
PP_0582	—	exclude (generic thiolase)	EC over-propagation
PP_2215	—	exclude (generic thiolase)	acetyl-CoA acetyltransferase
PP_3355	—	exclude (generic thiolase)	β-ketothiolase
PP_3754	bktB	exclude (generic thiolase)	β-ketothiolase
PP_4636	yqeF	exclude (generic thiolase)	acetyl-CoA acetyltransferase
PP_0158	gcdH	exclude (unrelated)	glutaryl-CoA DH
PP_1218	—	exclude (unrelated)	acyl-CoA thioesterase
PP_2504	—	exclude (meta-cleavage)	4-oxalocrotonate tautomerase
PP_1791	—	exclude (meta-cleavage)	4-OH-2-oxovalerate aldolase
PP_1950	—	exclude (unrelated)	cytochrome P450
PP_0847	fdx	exclude (unrelated)	2Fe-2S ferredoxin

Summary counts: 8 core enzymatic + 1 feeder + 3 transport/regulatory = **12 module members**; 4 catechol-branch (model separately); 4 ben upstream; **1 candidate_uncertain (PP_3648)**; **19 exclude (over-propagated/unrelated)**.
